Lecture 17 (10/23)

More examples of multiple-predictor regression

Midterm 2 material overview (Units 3 and 4)

Unit 3: estimation and single-predictor regression

- Estimators vs. parameters; Sample distribution.
- Bias, variance, MSE, consistency; standard errors and confidence intervals.
- Least-squares slope and intercept; Sample distribution of slope; R^2 and p-values (their relation to sample dist.)
- AR processes: stationarity, lagged regression, autocorrelation.

Unit 4: multiple-predictor regression

- Randomized trials vs. observational studies; confounding variables.
- Interpret coefficients in multiple predictor model
- Derive and apply omitted variable bias for one or many omitted predictors; Simpson's paradox.
- Categorical predictors and dummy variables.
- Effects on sample distribution of adding predictors.

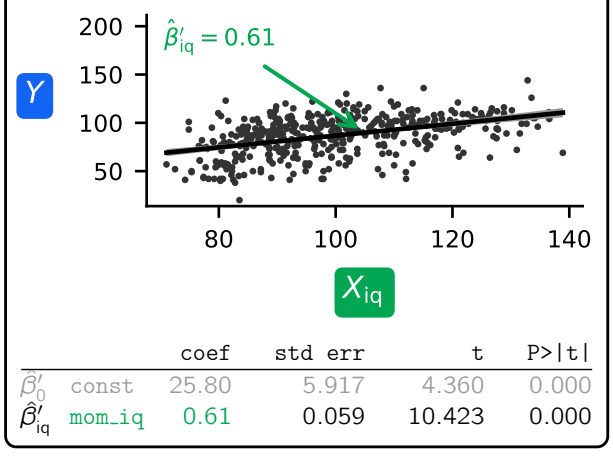
Today: Focus on applied examples of the two-predictor model.

Monday: Focus on derivations.

i	Y_i	$X_{\text{iq},i}$	$X_{\text{hs},i}$
	kid_score	mom_iq	mom_hs
1	65	121.1	1
2	98	89.4	1
⋮	⋮	⋮	⋮
6	98	107.9	0
⋮	⋮	⋮	⋮
N	70	91.3	1

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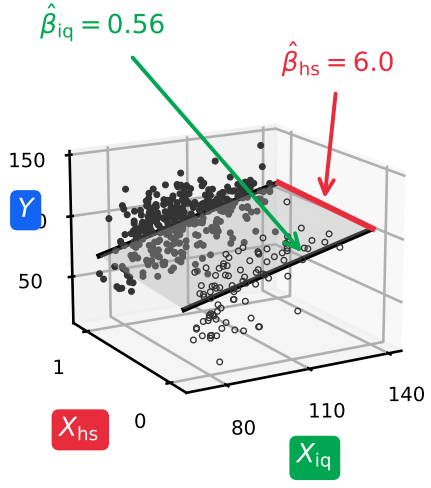
$$Y = \beta'_0 + \beta'_{\text{iq}} X_{\text{iq}} + \epsilon'$$

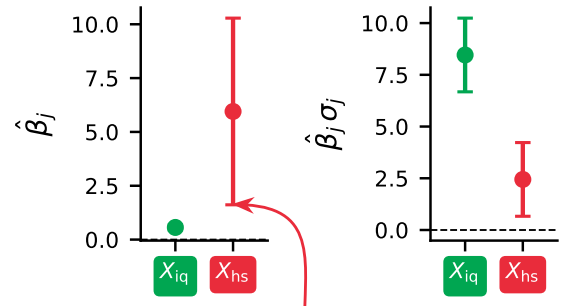


		coef	std err	t	P>|t|
$\hat{\beta}'_0$	const	25.80	5.917	4.360	0.000
$\hat{\beta}'_{\text{iq}}$	mom_iq	0.61	0.059	10.423	0.000

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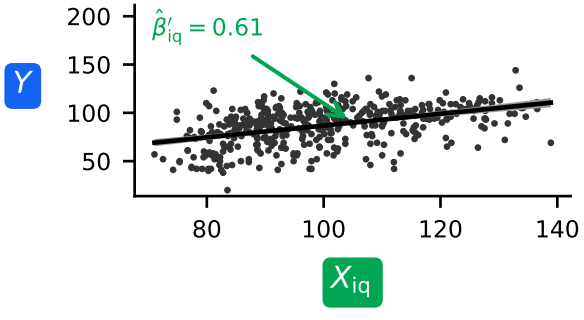
$$Y = \beta_0 + \beta_{\text{iq}} X_{\text{iq}} + \beta_{\text{hs}} X_{\text{hs}} + \epsilon$$





		coef $\hat{\theta}$	std err se($\hat{\theta}$)	t $\hat{T}$	P>|t| p_v
$\hat{\beta}_0$	const	25.73	5.875	4.380	0.000
$\hat{\beta}_{\text{iq}}$	mom_iq	0.56	0.061	9.309	0.000
$\hat{\beta}_{\text{hs}}$	mom_hs	5.95	2.212	2.690	0.007

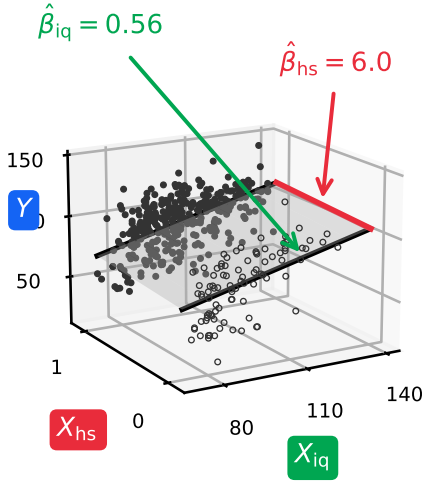
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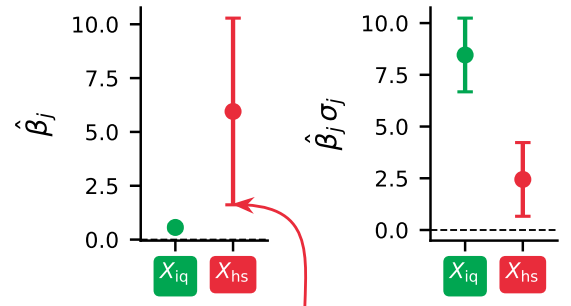


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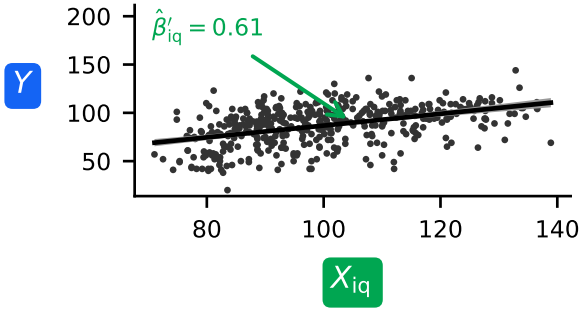
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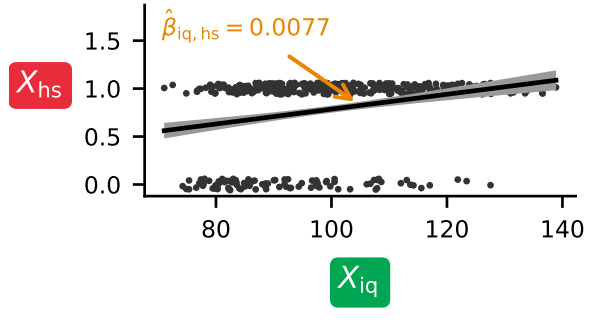
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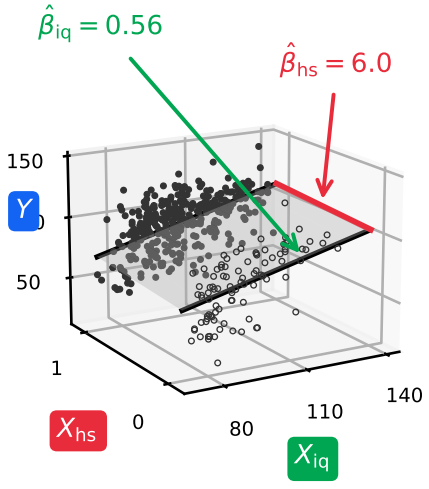
$$X_{\text{hs}} = \gamma_0 + \beta_{\text{iq,hs}} X_{\text{iq}} + \delta$$

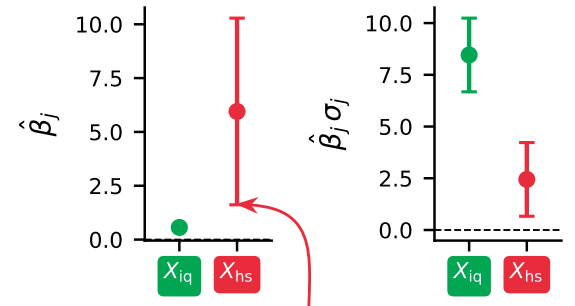


		coef	std err	t	P>|t|
$\hat{\gamma}_0$	const	0.0115	0.128	0.090	0.929
$\hat{\beta}_{\text{iq,hs}}$	mom_iq	0.0077	0.001	6.126	0.000

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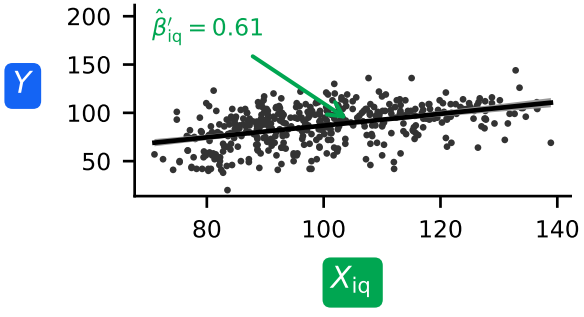
$$Y = \beta_0 + \beta_{\text{iq}} X_{\text{iq}} + \beta_{\text{hs}} X_{\text{hs}} + \epsilon$$





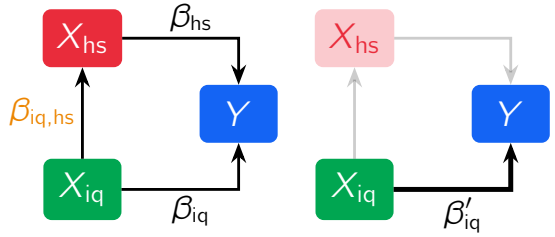
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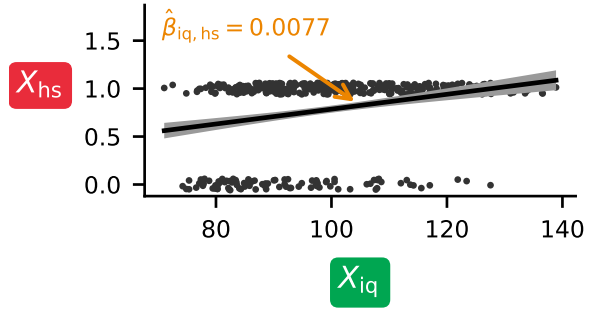
$$\beta'_{\text{iq}} = \beta_{\text{iq}} + \beta_{\text{hs}} \beta_{\text{iq,hs}}$$



both X_{hs} left out

omitted variable bias

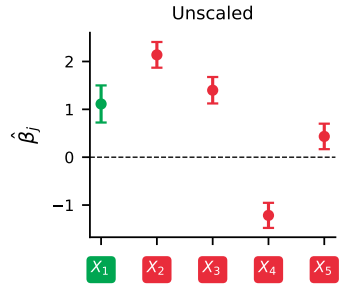
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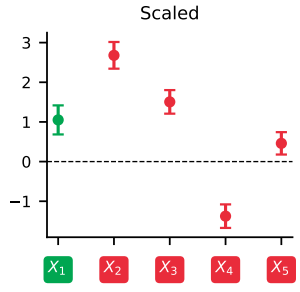


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i	Y_i	$X_{1,i}$	$X_{2,i}$	$X_{3,i}$	$X_{4,i}$	$X_{5,i}$
1	6.7	1.7	3.0	0.3	1.1	0.4
2	6.0	−0.3	1.5	0.7	0.8	−0.9
3	−4.7	−0.5	−2.0	0.3	−1.1	−1.4
⋮	⋮	⋮	⋮	⋮	⋮	⋮
N	6.6	0.4	0.8	0.2	−1.9	1.0

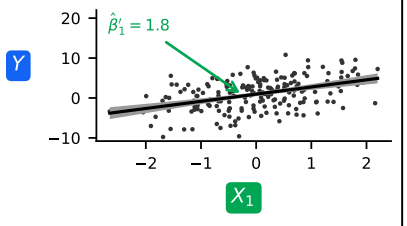
$$Y = \beta_0 + \beta_1 X_1 + \sum_{j=2}^{5} \beta_j X_j + \epsilon$$





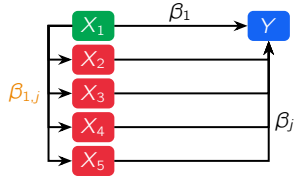
		coef $\hat{\theta}$	std err se($\hat{\theta}$)	t $\hat{T}$	P>|t| p_v
$\hat{\beta}_0$	const	1.01	0.137	7.370	0.000
$\hat{\beta}_1$	x1	1.11	0.198	5.622	0.000
$\hat{\beta}_2$	x2	2.14	0.137	15.604	0.000
$\hat{\beta}_3$	x3	1.40	0.141	9.925	0.000
$\hat{\beta}_4$	x4	−1.22	0.134	−9.090	0.000
$\hat{\beta}_5$	x5	0.43	0.136	3.195	0.002

$$Y = \beta'_0 + \beta'_1 X_1 + \epsilon'$$

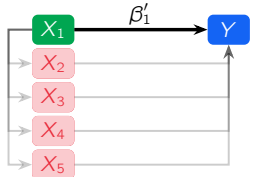


		coef	std err	t	P>|t|
$\hat{\beta}'_0$	const	0.96	0.251	3.807	0.000
$\hat{\beta}'_1$	x1	1.79	0.267	6.731	0.000

$$\beta'_1 = \beta_1 + \sum_{j=2}^{5} \beta_j \beta_{1,j}$$



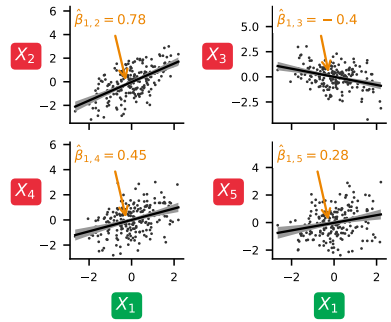
all predictors



$X_2, \ldots, X_5$ left out

omitted variable bias

$$X_j = \gamma_{0j} + \beta_{1,j} X_1 + \delta_j$$



		coef	std err	t	P>|t|
$\hat{\beta}_{1,2}$	x1	0.78	0.076	10.333	0.000
$\hat{\beta}_{1,3}$	x1	−0.40	0.076	−5.331	0.000
$\hat{\beta}_{1,4}$	x1	0.45	0.079	5.759	0.000
$\hat{\beta}_{1,5}$	x1	0.28	0.077	3.574	0.000

Example

Which hospital is better? A study follows 1000 patients, 500 at each hospital. Y is days until recovery. X_A is an indicator for hospital (1 if A, 0 if B). X_{sev} indicates severity (1 for a severe case, 0 otherwise).

Dep. Variable:		days	R-squared:		0.09
	coef	std err	t	P>|t|	[0.025	0.975]
const	7	0.21	32.97	0.00	6.58	7.42
hosp_A	3	0.30	9.99	0.00	2.41	3.59

Within either hospital, severe cases take 10 days longer to recover on average than mild cases. Severe cases make up 70% of hospital A's patients and 20% of hospital B's patients.

Question

Which hospital would you choose?

Solution: comparing hospitals

1) The unadjusted line is $E[Y \mid X_A] = 7 + \boxed{3}\, X_A$. The unadjusted difference is 3 days.

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2) From the supplied information, the severity coefficient is $\boxed{\beta_{\text{sev}} = 10}$. The auxiliary slope is the difference in severe-case proportions:

$$\boxed{\beta_{A,\text{sev}}} = E[X_{\text{sev}} \mid X_A = 1] - E[X_{\text{sev}} \mid X_A = 0] = 0.70 - 0.20 = \boxed{0.50}.$$

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3) Apply OVB and solve for the adjusted hospital coefficient:

$$\boxed{\beta'_A} = \beta_A + \boxed{\beta_{\text{sev}}}\,\boxed{\beta_{A,\text{sev}}}, \qquad 3 = \beta_A + 10(0.5)$$

$$\boxed{\beta_A = 3 - 5 = -2}.$$

Solution: comparing hospitals

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4) Adjusted difference: -2 days. Choose A: predicted recovery is two days shorter at either severity.

Example

Is blood pressure related to age? Group A samples 400 people; Group B samples 1500 from the same population. Y is systolic blood pressure (mmHg), and X_{age} is age in decades. Only Group B also measures X_{bmi}, which is body mass index.

Group A

Dep. Variable:			bp	R-squared:		
	coef	std err	t	P>|t|	[0.025	0.975]
const	124	3.22	38.55	0.00	117.68	130.32
age	7	0.62	11.36	0.00	5.79	8.21

Group B

Dep. Variable:			bp	R-squared:		
	coef	std err	t	P>|t|	[0.025	0.975]
const	90	1.62	55.42	0.00	86.81	93.19
age	3	0.27	11.12	0.00	2.47	3.53
bmi	2	0.06	33.16	0.00	1.88	2.12
Dep. Variable:			bmi	R-squared:		
	coef	std err	t	P>|t|	[0.025	0.975]
const	17	0.54	31.51	0.00	15.94	18.06
age	2	0.10	19.35	0.00	1.80	2.20
Dep. Variable:			age	R-squared:		
	coef	std err	t	P>|t|	[0.025	0.975]
const	2.3	0.14	16.00	0.00	2.02	2.58
bmi	0.1	0.01	19.35	0.00	0.09	0.11

Question

Are the results of the two groups consistent?

Solution: blood pressure and omitted BMI

1) Group B estimates the adjusted relationship:

$$E[Y \mid X_{\text{age}}, X_{\text{bmi}}] = 90 + \boxed{3}\, X_{\text{age}} + \boxed{2}\, X_{\text{bmi}}.$$

Its auxiliary regression gives $E[X_{\text{bmi}} \mid X_{\text{age}}] = 17 + \boxed{2}\, X_{\text{age}}$.

Solution: blood pressure and omitted BMI

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2) Omitting BMI changes the age slope by the product of two slopes:

$$\boxed{\hat{\beta}'_{\text{age}}} = \boxed{3} + \boxed{2} \times \boxed{2} = \boxed{7}.$$

   That agrees with Group A's fitted age slope.

Solution: blood pressure and omitted BMI

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3) The intercept also changes:

$$\hat{\beta}'_0 = 90 + 2 \times 17 = 124.$$

   Thus Group B implies the reduced line $E[Y \mid X_{\text{age}}] = 124 + 7X_{\text{age}}$, exactly the reported Group A line (at the displayed precision).

Solution: blood pressure and omitted BMI

1) Group B estimates the adjusted relationship:

$$E[Y \mid X_{\text{age}}, X_{\text{bmi}}] = 90 + \boxed{3}\,\boxed{X_{\text{age}}} + \boxed{2}\,\boxed{X_{\text{bmi}}}.$$

   Its auxiliary regression gives $E[X_{\text{bmi}} \mid X_{\text{age}}] = 17 + \boxed{2}\,\boxed{X_{\text{age}}}$.

2) Omitting BMI changes the age slope by the product of two slopes:

$$\boxed{\hat{\beta}'_{\text{age}}} = \boxed{3} + \boxed{2} \times \boxed{2} = \boxed{7}.$$

   That agrees with Group A's fitted age slope.

3) The intercept also changes:

$$\hat{\beta}'_0 = 90 + 2 \times 17 = 124.$$

   Thus Group B implies the reduced line $E[Y \mid X_{\text{age}}] = 124 + 7X_{\text{age}}$, exactly the reported Group A line (at the displayed precision).

4) Compare with Group A: slope $7 = 7$ and intercept $124 = 124$. The fitted coefficients agree.

Example

Do books at home raise children's reading scores? A survey includes 600 children. Y is a reading test score, and X_{books} is the number of books at home, measured in tens.

Dep. Variable:		score	R-squared:		0.51
	coef	std err	t	P>|t|	[0.025	0.975]
const	42	2.04	20.57	0.00	37.99	46.01
books	8	0.32	24.78	0.00	7.37	8.63

Two possible confounders are X_{educ}, parent's years of education, and X_{inc}, household income in \$10k. Assume the following linear regression relationships hold in this population:

- among children with the same number of books and the same income, each extra year of parent's education goes with 2 more points on average;
- among children with the same number of books and the same parent's education, each extra \$10k of income goes with 1 more point on average;
- each extra 10 books goes with 1 more year of parent's education and \$20k more income on average.

Question

What is the coefficient on `books` in a model that also includes X_{educ} and X_{inc}?

Solution: books, education, and income

1) Translate the given effects into full-model coefficients:

$$E[Y \mid X_{\text{books}}, X_{\text{educ}}, X_{\text{inc}}] = \hat{\beta}_0 + \hat{\beta}_{\text{books}} X_{\text{books}} + \boxed{2} X_{\text{educ}} + \boxed{1} X_{\text{inc}}.$$

Income is measured in units of \$10,000.

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Income is measured in units of \$10,000.

2) Each additional ten books corresponds to one year of education and two income units, so the auxiliary slopes are

$$\boxed{\beta_{\text{books,educ}} = 1}, \qquad \boxed{\beta_{\text{books,inc}} = 2}.$$

Solution: books, education, and income

1) Translate the given effects into full-model coefficients:

$$E[Y \mid X_{\text{books}}, X_{\text{educ}}, X_{\text{inc}}] = \hat{\beta}_0 + \hat{\beta}_{\text{books}} X_{\text{books}} + \boxed{2} X_{\text{educ}} + \boxed{1} X_{\text{inc}}.$$

Income is measured in units of \$10,000.

2) Each additional ten books corresponds to one year of education and two income units, so the auxiliary slopes are

$$\boxed{\beta_{\text{books,educ}} = 1}, \qquad \boxed{\beta_{\text{books,inc}} = 2}.$$

3) Apply OVB with both omitted predictors:

$$\boxed{8} = \hat{\beta}_{\text{books}} + \boxed{2} \times \boxed{1} + \boxed{1} \times \boxed{2}.$$

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4) Solve: $\hat{\beta}_{\text{books}} = 4$.

Example

Does a new training program improve athletic performance? In an observational study, athletes who joined the program scored 6 points higher on average on a fitness test than athletes who did not join.

Y is the fitness score. X_{program} is an indicator for participation (1 for participants, 0 for nonparticipants). The groups also differed in their habits before the program began:

Average before the program	Participants	Nonparticipants
Sleep (hours per night)	8	7
Exercise (hours per week)	5	3

In a linear model including program participation, prior sleep, and prior exercise, the coefficients on sleep and exercise are 2 points per hour and 1 point per hour, respectively.

Question

What is the coefficient on program participation after controlling for prior sleep and exercise?

Solution: the training-program comparison

1) The raw difference is the single-predictor coefficient:

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3) The supplied full-model coefficients are 2 for sleep and 1 for exercise. Apply OVB:

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4) Solve: $\boxed{\beta_{\text{program}} = 6 - 2 - 2 = 2}$ points.

Example

What makes a wildfire spread? A fictional study includes 25 wildfires. Y is area burned in acres. X_{drought} is a drought index from 0 to 100, and X_{wind} is wind speed in mph.

Dep. Variable:		acres	R-squared:			0.80
	coef	std err	t	P>|t|	[0.025	0.975]
const	-220	163.46	-1.35	0.19	-558.99	118.99
drought	5	3.73	1.34	0.19	-2.74	12.74
wind	5	9.14	0.55	0.59	-13.95	23.95
Dep. Variable:		acres	R-squared:			0.80
	coef	std err	t	P>|t|	[0.025	0.975]
const	-305	50.01	-6.10	0.00	-408.46	-201.54
drought	7	0.73	9.53	0.00	5.48	8.52
Dep. Variable:		acres	R-squared:			0.78
	coef	std err	t	P>|t|	[0.025	0.975]
const	-2.4	19.10	-0.13	0.90	-41.91	37.11
wind	17	1.86	9.14	0.00	13.15	20.85

Question

Is area burned related to drought? To wind? Do the large p-values in the first model mean that neither matters?

Solution: wildfire predictors and uncertainty

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3) Use the changes in slope to infer the auxiliary slopes:

$$7 = 5 + 5\beta_{\text{drought,wind}} \Rightarrow \boxed{\beta_{\text{drought,wind}} = 0.4},$$

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4) Their product is $\rho^2 = 0.4(2.4) = 0.96$, so $\rho \approx 0.98$. High collinearity explains the large joint-model SEs; the large p-values do not imply zero coefficients.